Notes on Characteristic Classes and its Application in Physics

Yanpu Wu

June 15, 2025

0 Introduction

In this survey, we will mainly focus on characteristic class, including its definition, calculation and some application in physics.

We would discuss some physical aspects first, without rigorous definition of many necessary machinery used. Nevertheless, I hope this part would exhibit the wonderful intuition and idea that physicists had to explain many seemingly weird physical phenomena and to open the door of the quantum world by generalizing the observations. The definition, notation and conclusions that appear in this section can be found in the later sections.

After this physical introduction and background, we would set up the mathematical foundation. The most fundamental thing is the characterstic classes. In the course, we defined some characteristic classes such as Chern classes directly by the curvature form. Nevertheless there is a more axiomatic way to define them, we shall roughly present these axioms without proving the existence and uniqueness under this definition in detail.

However, we shall return to the definition by curvature form in the later sections

and review it as a result from Chern-Weil theory. Before that we need some preparations about connections and curvature forms, especially proving the important Bianchi identity used in the proof. Also we would see some mathematical applications of them and these would together serve as the mathematical background for physics that we desired for.

1 Story from Physics

We will mainly focus on two physical aspects in this section. The first is about monopoles. We shall first discuss the classical monopoles found by Dirac and explain how he argued that the electric charge of all particles must be an integral multiple of a certain fundamental unit by the existence of such a monopole. Next, we shall introduce the generalization of Dirac monopole to the Yang-Mills field, i.e., the Yang monopole. The second is what we've discussed in the class: the instantons. We shall see how the mathematical objects defined below would be applied in the setting of physics. (So "recall" in this chapter really means see below.)

1.1 Monopoles

First let's disucss the Dirac monopoles. Now the setting is a principal $U(1)-$bundle P over $\mathbb{R}^3 - \{0\} \simeq \mathbb{S}^2$. In fact, in this case $c_1(\mathbf{F}) = \mathrm{tr}(\frac{i\mathbf{F}}{2\pi}) = n \in \mathbb{Z}$ (and this can also be used to prove $\pi_1(U(1)) = \pi_1(\mathbb{S}^1) = \mathbb{Z}$ as before), which is defined to be the *magnetic charge*.

Now we are indeed considering a electric-magnetic field and the curvature form $\mathbf{F}$ is its curvature. Consider the magnetic strength $\vec{B}$, then $\mathrm{div}\vec{B} = 0$ has a solution in $\mathbb{R}^3 - \{0\}$. The homology group of $\mathbb{R}^3 - \{0\}$ is not 0 hence there exists non-exact closed form, namely, which is not in the form $\vec{B} = \nabla \times \vec{A}$. In fact, we know that $H^2(\mathbb{R}^3 - \{0\}, \mathbb{R}) = \mathbb{R}$. Hence there exists B satisfying the Maxwell equations but is not

d of any potential, we call this kind of field a *magnetic monopole* or *Dirac monopole*. Note that unlike in usual $\mathbb{R}^3$, where any closed B (namely, satisfying the Maxwell equation $dB = 0$) is exact, hence its integral over an embedded sphere $\mathbb{S}^2$ is 0, for a monopole, this inetgral will be equal to m, the magnetic charge defined before.

Dirac is the first person who studied monopoles seriously, and he found that if there really exists one monopole, then we may deduce that all free particles have charges that are an integer multiple of the electron charge. Next we shall present his argument.

Recall that if we "drag" a particle with electric charge q around a loop γ that is a boundary of a 2-disk D, its wavefuction is multiplied by a phase $e^{-\frac{i}{\hbar}q\int_D B}$. And we see that if there exists a potential A such that $dA = B$, then $\int_D B = \int_\gamma A$ by Stokes formula hence this phase is independent from the disk D. However, now there doesn't exist such a potential. But if we insist the different ways of calculation caused by different D chosen would imply the same result, we may get Dirac's result.

Indeed, if we consider the loop γ that is the equator of a sphere, and let D_1 be the northern hemisphere and D_2 be the southern hemisphere. Then different orientation would give the following calculation result:

$$\int_{D_1} B = \frac{m}{4\pi}\int_0^{\frac{\pi}{2}}\int_0^{2\pi} \sin\phi d\theta \wedge d\phi = \frac{m}{2}$$
$$\int_{D_2} B = -\frac{m}{4\pi}\int_{\frac{\pi}{2}}^{\pi}\int_0^{2\pi} \sin\phi d\theta \wedge d\phi = -\frac{m}{2}.$$

Hence they would each give a phase of $e^{-\frac{iqm}{2\hbar}}$ and $e^{\frac{iqm}{2\hbar}}$. If these two are equal then we must have $e^{\frac{iqm}{\hbar}} = 1$. Namely, q must be an integer multiple of $\frac{2\pi\hbar}{m}$. Recall that $\hbar$ is defined to be $\frac{h}{2\pi}$, then this just means $q = \frac{Nh}{m}$ for some integer N and the claim is proved.

Via Chern-Weil theory, we might understand the Dirac monopole to be a $U(1)-$principal bundle with non-trivial first Chern class on a spacetime of the form $(\mathbb{R}^3_{\text{space}} - \{0\}) \times$

$\mathbb{R}_{\text{time}}$, witnessed by the magnetic charge $m = \int_{\mathbb{S}^2} F_\nabla \in \mathbb{Z} \subseteq \mathbb{R}$, where F_∇ is the curvature 2-form of any principal connection ∇ on this bundle.

In this way, we might easily generalize to the case over a Yang-Mills field. Namely, a *Yang Monopole* is just a $SU(2)-$principal bundle with non-trivial second Chern class on a spacetime of the form $(\mathbb{R}^5_{\text{space}} - \{0\}) \times \mathbb{R}_{\text{time}}$, witnessed by the instanton number $m = \int_{\mathbb{S}^4} F_\nabla \wedge F_\nabla \in \mathbb{Z} \subseteq \mathbb{R}$, where F_∇ is the curvature 2-form of any principal connection ∇ on this bundle.

1.2 Instantons

In the discussion about Yang monopoles above we saw the instanton number. So we shall now look at the example of instantons. Recall that when the structure group of the bundle is $SU(2)$, the first two Chern classes are $c_1(P) = 0, c_2(P) = \frac{1}{8\pi^2}\text{tr}(\mathbf{F} \wedge \mathbf{F})$. By physical abuse of notation, we sometimes write $c_i(P)$ as $c_i(\mathbf{F})$ and sometimes write $\frac{1}{8\pi^2}\text{tr}(\mathbf{F} \wedge \mathbf{F}) = -\frac{1}{16\pi^2}\text{tr}(\mathbf{F}^a \wedge \mathbf{F}^a)$.

We denote $-k = c_2(P) = \frac{1}{8\pi^2}\int_{\mathbb{S}^4} \text{tr}(\mathbf{F} \wedge \mathbf{F})$, the second Chern number, and k is in fact the so-called *instanton number*. Consider $\mathbf{F} = \mathbf{F}^+ + \mathbf{F}^-$, where $\mathbf{F}^+ = \frac{1}{2}(\mathbf{F} + *\mathbf{F})$ and $\mathbf{F}^- = \frac{1}{2}(\mathbf{F} - *\mathbf{F})$, in which $*$ is the Hodge operator.

Note that $\text{tr}(A \wedge B) = \text{tr}(B \wedge A)$ and $*\mathbf{F}^+ = \mathbf{F}^+, *\mathbf{F}^- = -\mathbf{F}^+$. Now we first use this obeservation to calculate the action

$$\begin{aligned} S &= -\int_{\mathbb{S}^4} \text{tr}(\mathbf{F} \wedge *\mathbf{F}) = \text{tr}\int_{\mathbb{S}^4} (\mathbf{F}^+ + \mathbf{F}^-) \wedge *(\mathbf{F}^+ + \mathbf{F}^-) \\ &= -\text{tr}\int_{\mathbb{S}^4} \mathbf{F}^+ \wedge \mathbf{F}^+ + \text{tr}\int_{\mathbb{S}^4} \mathbf{F}^- \wedge \mathbf{F}^-. \end{aligned}$$

We consider the norm defined by $\|w\| := -\text{tr}\langle w, w\rangle = \text{tr}\int w \wedge *w$. With this notation, we immediately see that $S = \|\mathbf{F}^+\|^2 + \|\mathbf{F}^-\|^2$ and in particular it is positive.

Similarly we can calculate $8\pi^2 k = -\int_{\mathbb{S}^4} \text{tr}(\mathbf{F} \wedge \mathbf{F})$. What is different from the calculation above is that the cross term involving $\langle \mathbf{F}^+, \mathbf{F}^-\rangle$ still exists, but in fact we could see this is 0 using the symmetry of the inner product (since it is defined by such

an integral), i.e., this decomposition is an orthogonal one. By this observation we see that

$$8\pi^2 k = \|\mathbf{F}^+\|^2 - \|\mathbf{F}^-\|^2.$$

Then we see in particular that $S \geq 8\pi^2|k|$. The connection $\mathbf{A}$ with respect to the curvature form $\mathbf{F}$ is called the *instanton*. If $k = S = 0$, then $\mathbf{F} = \mathbf{F}^+ = \mathbf{F}^- = 0$. In this case the instanton $A_\mu(x)$, which is a flat connection, is called the *gauge potential*. If $k > 0$ and $S = 8\pi^2 k$, we easily see that $\mathbf{F}^- = 0$ and $\mathbf{F} = \mathbf{F}^+ = *\mathbf{F}$, hence this solution is called *self-dual*. Similarly, if $k < 0$ and $S = -8\pi^2 k$, we see that $\mathbf{F} = - * \mathbf{F}$, hence this solution is called *anti-self-dual*.

Next we shall discuss the geometric meaning of the instanton number $k = -c_2(P)$. We will see this could be used to calculate the homotopy group $\pi_3(\mathbb{S}^3) = \pi_3(SU(2)) = \mathbb{Z}$. We can view $\mathbb{S}^3 \simeq SU(2)$ to be the boundary of $\mathbb{R}^4$. We usually find bond between some boundary relations and the topological behavior of a certain field and we shall now think in this manner.

We need some preparations on the obeservation that $\mathbb{S}^4$ is the one-point compactification of $\mathbb{R}^4$ via the sterographic projection. This is a conformal mapping and we need to use the fact that the metric tensor field as well as the Hodge $*-$operator, is conformally invariant. Hence the action $I = -\frac{1}{e^2}\int_M \mathrm{tr}(\mathbf{F} \wedge *\mathbf{F})$ is invariant. This allows us to define the $SU(2)-$gauge potential $A_\mu(x)$ on $\mathbb{S}^4$. Since $\mathbb{R}^4$ is contractible, any bundle over it is trivial but the case of its compactification $\mathbb{S}^4$ is different and hence there exists topologically non-trivial gauge field over $\mathbb{S}^4$. In fact, the curvature form is just its strength and the corresponding connection is the potential. We say that the instanton number marks the topological property of the $SU(2)-$principal bundle P over $\mathbb{S}^4$ and thus marks the non-trivial asymptotic property of the gauge potential $A_\mu(x)$.

Indeed, $c_2(P) \in H^4(\mathbb{S}^4, \mathbb{Z})$ is represented by a close 4-form over $\mathbb{S}^4$, hence when $k = c_2 \neq 0$, this closed form is not exact. Cover $\mathbb{S}^4$ by the two hemispheres S_1, S_2,

which are contractible. In fact, locally we may use the Chern-Simons formula to write explicityly:

$$\mathrm{tr}(\mathbf{F}\wedge\mathbf{F}) = d(\mathrm{tr}(\mathbf{F}\wedge\mathbf{A} - \frac{1}{3}\mathbf{A}\wedge\mathbf{A}\wedge\mathbf{A})).$$

In fact, on the overlap, the corresponding connection is connected with the gauge transformation $\mathbf{A}_1 = g^{-1}\mathbf{A}_2 g + g^{-1}dg$ and hence $\mathbf{F}_1 = g^{-1}\mathbf{F}_2 g$. Together with this observation, since $\mathbb{S}^3$ is the boundary of S_1, S_2 but with different orientation, by Stokes formula we have the following calculation

$$\begin{aligned}
-8\pi^2 k =& \mathrm{tr}\int_{\mathbb{S}^4}(\mathbf{F}\wedge\mathbf{F}) = \mathrm{tr}\int_{S_1}(\mathbf{F}\wedge\mathbf{F}) + \mathrm{tr}\int_{S_2}(\mathbf{F}\wedge\mathbf{F}) \\
&= \int_{\mathbb{S}^3}\mathrm{tr}(\mathbf{F}_1\wedge\mathbf{A}_1 - \frac{1}{3}\mathbf{A}_1\wedge\mathbf{A}_1\wedge\mathbf{A}_1) - \mathrm{tr}(\mathbf{F}_2\wedge\mathbf{A}_2 - \frac{1}{3}\mathbf{A}_2\wedge\mathbf{A}_2\wedge\mathbf{A}_2) \\
&= -\frac{1}{3}\int_{\mathbb{S}^3}(g^{-1}dg)\wedge(g^{-1}dg)\wedge(g^{-1}dg).
\end{aligned}$$

In summary, for $g \in SU(2)$ we have that

$$k = \frac{1}{24\pi^2}\int_{\mathbb{S}^3}(g^{-1}dg)\wedge(g^{-1}dg)\wedge(g^{-1}dg) \in \mathbb{Z}.$$

As the case in calculation of $\pi_1(\mathbb{S}^1)$ and $\pi_2(\mathbb{S}^2)$, we can prove that $\pi_3(\mathbb{S}^3) = \mathbb{Z}$ as well.

Next we shall construct a gauge field whose instanton number $k = 1$. Consider the gauge potential, i.e., the connection given by $\mathbf{A}(x) = \frac{1}{|x^2|+1}\Im(xd\bar{x})$, where $x \in \mathbb{R}^4$ is viewed as an element in $\mathbb{H}$ and its conjugate and norm is defined as a quaternion. The corresponding gauge field strength, i.e., the curvature is $\mathbf{F}(x) = \frac{dx\wedge d\bar{x}}{|x^2|+1}$. We could directly calculate $dx \wedge d\bar{x}$ and see it is in particular $\Im\mathbb{H}-$valued. By the following calculation, we indeed see the instanton number of this gauge field is 1, where we

denote $dx_{ij} = dx_i \wedge dx_j$:

$$\begin{aligned}
k =& \frac{1}{4\pi^2}\int_{\mathbb{R}^4} |\mathbf{F}|^2 (dx)^4 \\
=& \frac{1}{4\pi^2}\int_{\mathbb{R}^4} \frac{4}{(|x|^2+1)^4} |-(dx_{12}+dx_{34})\mathbf{i} - (dx_{13}-dx_{24})\mathbf{j} - (dx_{14}+dx_{23})\mathbf{k}| (dx)^4 \\
=& \frac{1}{4\pi^2}\int_{\mathbb{R}^4} |\frac{4}{(|x|^2+1)^4}(|(dx_{12}+dx_{34})|^2 + |(dx_{13}-dx_{24})|^2 + |(dx_{14}+dx_{23})|^2)(dx)^4 \\
=& \frac{1}{4\pi^2}\int_{\mathbb{R}^4} \frac{24}{(|x|^2+1)^4} = 1,
\end{aligned}$$

where we've used $|dx_{12} + dx_{34}|^2 = 2$ and the calculation is done.

2 Characteristic Classes

2.1 Basic Idea

We start from the definition of the so-called universal bundle:

Def 2.1. The $(n-k-1)-$*universal bundle* $\xi(n-k-1, O(k))$ is defined as follows: the fiber and the structure group is $O(k)$ and the base space is the coset space $O(n)/O(k)\times O(n-k)$, i.e., the Grassmannian $\mathrm{Gr}(n,k,\mathbb{R})$, consisting of the $k-$dim subspaces in $\mathbb{R}^n$. (Namely, the points in $\mathrm{Gr}(n,k,\mathbb{R})$ are these $k-$dim subspaces in $\mathbb{R}^n$.)

E.g. 2.1. For $n = 3, k = 1$, the Grassmannian is just $\mathbb{R}P^2$.

Rmk 2.1. (1) For historical remark, after Grassmann first came up with this idea, no one understood him so he turned to study the Sanskrit language. It has not been given much notice until the 1950s', when so many mathematicians like Todd, Serre, Milnor, Borel, Hirzebruch did calculation on its topology, especially its homology, for example, in their doctoral theses.

(2) The total space of the universal bundle is just $O(n)/O(n-k)$, with the structure group $O(k)$. It is also known as the Stiefel manifold for the $k-$frame in $\mathbb{R}^n$. It is a

compact homogeneous space. As its non-compact analogue, we may consider the upper half plane $\mathbb{H} = \mathrm{SL}(2,\mathbb{R})/O(2) = O(2,1)/O(2)\times O(1)$.

The name "universal" comes from the following essential result in topology that we present here without proof:

Prop 2.1. If P is a principal bundle on a $d-$dimensional manifold M with structure group $O(k)$, then we can always for find n sufficiently large, a map $f : M \to \mathrm{Gr}(n,k,\mathbb{R})$, such that $f^*\xi(n-k-1,O(k)) = P$.

This means that we have a way to compare any given principal bundle P with structure group $O(k)$ to the universal bundle. In fact this is the idea of Понтрягин: to compare any principal bundle P and ξ is done by pulling-back bundles. More concretely, if there is a map $f : M \to \mathrm{Gr}(n,k,\mathbb{R})$, then we could pull back a bundle on the grassmannian to a bundle on M, namely, $f^*\xi(n-k-1,O(k))$. This is indeed a principal bundle on M with the same structure group $O(k)$ with the universal bundle $\xi(n-k-1,O(k))$.

Rmk 2.2. Let's discuss why the integer $n-k-1$ is useful in the definition of the universal bundle $\xi(n-k-1,O(k))$. In fact, $\pi_p(O(n)/O(n-k)) = 0, 0 \le p \le n-k-1$. In other words, it is $(n-k-1)-$connected.

Also, for principal bundles with structure group $O(k)$ over a manifold whose dimension $\le n-k-1$, it can always be classified by $\xi(n-k-1,O(k))$. In fact, if we fix n,k, then for any varying $f : M \to \mathrm{Gr}(n,k,\mathbb{R})$, $f^*\xi(n-k-1,O(k))$ gives every principal bundle P with group $O(k)$ on a manifold whose dimension $d \le n-k-1$. This is because the sufficient large n in Prop 1.1. shall at least satisfy that $n \ge d+k+1$.

For another geometric explanation, since the base space $\mathrm{Gr}(n,k,\mathbb{R})$ carries a natural action on the $k-$subspaces by the group $O(k)$, giving the fiber of $\xi(n-k-1,O(k))$ and the structure group $O(k)$. We know a trivial principal bundle with base M and

group G can be expressed as $M \times G$, hence the nontriviality of the universal bundle $\xi(n-k-1, O(k))$ can be viewed as that to what extent $\mathrm{Gr}(n,k,\mathbb{R}) \times O(k) = \{O(n)/O(n-k) \times O(k)\} \times O(k)$ is different from $O(n)/O(n-k)$.

Rmk 2.3. For historical remark, Понтрягин was the first to study characterstic classes. His teacher Alexanderoff is a Russian topologist who studied the topology on the sphere, while Понтрягин find that this can be generalized to the case of Grassmannian and found the characterstic classes in 1930s'.

In Понтрягин's opinion, there are two kind of mathematical problems. What exemplifies the first is the Goldbach conjecture, while what exemplifies the second is the differential dynamic system theory by Poincaré. He thought that the second one is more interesting to his own taste, since it is all about delving into the unknown and the mysterious world, while the answer to a conjecture is simply yes or no.

A natural question to ask next is that what if the group here is not $O(k)$ or something as $U(k), Sp(k)$? This is also an insight from Понтрягин. The final answer is that it can be applied to any connected Lie group. Next we will show how this is realized.

Thm 2.1. If G is a compact group, then G is isomorphic to a subgroup of $O(m)$ for sufficiently big m.

Proof. Set $n = \dim G$, consider a faithful representation of G on a real vector space V with dimension m. This is obtained as follow. For any positive definite inner product $(,)$ on V, hence by averaging we can obtain a $G-$invariant inner product which is also real positive definite. Then we can obtain a faithful representation by orthogonal matrices, namely, G is isomorphic to a subgroup of $O(m)$. □

E.g. 2.2. $U(n)$ is isomorphic to a subgroup of $O(2n)$.

Then for a given compact group G, we have a universal bundle $\xi(n-k-1, G)$.

Finally, even when G is not a compact group but any connected Lie group, we can find a maximal compact subgroup H with a decomposition $G = H \times D$ with D contractible. Hence P can be viewed as a bundle with structure group H and we again reduce to the compact case, namely, we have a $(n-k-1)-$universal bundle $\xi(n-k-1, H)$ for a principal bundle P with connected Lie group G and maximal compact subgroup H.

Rmk 2.4. For $G = SO(k)$, we can substitute $O(n)$ with $SO(n)$, with the corresponding universal bundle denoted as $\tilde{\xi}$, whose base space is $SO(n)/SO(k) \times SO(n-k)$ and is an orientable Grassmannian manifold, denoted as $\widetilde{\mathrm{Gr}}(n, k, \mathbb{R})$.

For $G = U(k)$, we can substitute $O(n)$ with $U(n)$, with the corresponding universal bundle denoted as ξ_c, whose base space is $U(n)/U(k) \times U(n-k)$ and is a complex Grassmannian manifold, denoted as $\mathrm{Gr}(n, k, \mathbb{C})$. The real dimension of this manifold is $n^2 - k^2 - (n-k)^2 = 2k(n-k)$, since $\dim_{\mathbb{R}} U(n) = n^2$. Hence $\dim \mathrm{Gr}(n, k, \mathbb{C}) = 2\mathrm{Gr}(n, k, \mathbb{R})$. For another point of view, since $2k(n-k) = k(2n-2k)$, it can also be viewed as a $(2n-2k)$-universal bundle, and not only a $(n-k-1)$-universal bundle. Namely $\xi_c \equiv \xi_c(2n-2k, U(n))$.

2.2 Definition for Characterstic Classes

We're now close to the concept of charactestic classes. A motivation is to ask how many (equivalence classes of) principal bundle P with a given structure group G are there over a base M. We know from above that every such bundle is a pullback bundle $f^*\xi$ of a certain $f : M \to O(n)/G \times O(n-k)$. And notice that if $f \simeq g : M \to \mathrm{Gr}(n, k, \mathbb{R})$, then $f^*\xi \simeq g^*\xi$, i.e., homotopic maps give equivalent bundles. Thus we get an one-to-one correspondence between the equivalence classes of principal bundle P with a given structure group G, with the homotopic classes of maps from M to $O(n)/G \times O(n-k)$.

But both are very hard to calculate. So we want to try an easier problem: to set

up a criterion to determine how far is a bundle from a trivial bundle. It is a map c, revealing how nontrivial $\xi(n-k-1, G)$ is, and pullback it to P. It shall at least satisfy the following basic properties:

(1) If $P \simeq P'$, then $c(P) = c(P')$;

(2) $f^* c(\xi) = c(f^* \xi)$.

Let's first recall the cohomology groups to have some basic clues to construct such c. We notice that if $c(P) \in H^i(M, \mathbb{R})$ where P is a bundle on R, then c satisfies the properties above. This is because homotopic maps give the same cohomology class and d commutes with any pullback α^*.

Such object $c(\xi) \in H^i(B, \mathbb{R})$ is called *characterstic class*, where B is the base space. There are two steps in the calculation, the first is to try to calculate it for the universal bundle, and then pull it back to the original principal bundle P.

(1) For principal bundle with structure group $G = O(k)$, it is called *Pontrjagin class*.

(2) For principal bundle with structure group $G = SO(k)$, it is still *Pontrjagin class*, but there is also another more, namely, the *Euler class*, which is nontrivial only when k is even.

(3) For principal bundle with structure group $G = U(k)$, it is called *Chern class*.

(4) For principal bundle with structure group $G = O(k)$, but the cohomology is computed with coefficient $\mathbb{Z}/2\mathbb{Z}$, it is called *Stiefel-Whitney class*.

In fact, usually we demand more for these classes and the definition above is just a part of the full definition. We can define directly using the so-called Chern-Weil theory below, however we shall present an axiomatic way of definition here. We shall only give such definition for Chern classes as an example. (The other classes have similar definition as well.)

Def 2.2. Let $\xi = (E, \pi, B)$ be a complex $n-$plane bundle. The Chern class $c_i \in$

$H^{2i}(B,\mathbb{Z})$ is defined to satisfy the following axioms:

(1) $c_0(\xi) = 1, c_i(\xi) = 0$ for any $i > n$;

(2) (Naturality) If $f : B \to B'$ is covered by a bundle map from ξ to another complex $n-$plane bundle ξ', then $c_i(\xi) = f^* c_i(\xi')$;

(3) (Sum Formula) For two complex vector bundles ξ_1, ξ_2 over a paracompact base space B, we have that $c(\xi_1 \oplus \xi_2) = c(\xi_1)c(\xi_2)$, where $c(\xi) := \sum_{i=1}^{n} c_i(\xi)$ is the total Chern-class.

(4) For the universal bundle γ^1 over $G_1 = \mathbb{C}P^\infty$, we have that $c_1(\gamma^1) = e((\gamma^1)_{\mathbb{R}})$, where e is the Euler class.

Not being the main point of this survey paper, the proof of existence and uniqueness will be omitted.

The Stiefel-Whitney class is somehow special since the coefficient is $\mathbb{Z}/2\mathbb{Z}$, but the elements in rest 3 classes can be given by some real non-exact closed differential forms. So we will only present some basic properties of Stiefel-Whitney classes W_n here. The first fact is that when the dimension of the manifold M is even, then $W_n(E) = e(E)$. Also, we have the following properties that reveal the importance of Stiefel-Whitney classes (of small order):

Prop 2.2. (1) $W_1(TM) = 0$ if and only if M is orientable;

(2) If $W_2(TM) \neq 0$ then there doesn't exist spin structure on M.

Rmk 2.5. In mathematics, spin structure involves the spinor groups $\mathrm{Spin}(k)$ from orthogonal groups. We shall not discuss it in detail here.

In fact, for projective space we have $W_1(T\mathbb{C}P^n) = 0$ and $W_2(T\mathbb{C}P^n) = \begin{cases} 0 & n \text{ odd} \\ x & n \text{ even} \end{cases}$,

where x is the generator of $H^2(\mathbb{C}P^n, \mathbb{Z}/2\mathbb{Z})$. Hence the previous result in particular tells us there does not exist spin structure on $\mathbb{C}P^{2n}$.

The next natural problem is to calculate the characterstic classes, we shall only introduce some basic methods (for Euler classes, Понтрягин classes and Chern classes), using connection and curvature as in the differential geometry in the following sections, which will also involve some concepts in general relativity theory and gauge fields. (In fact, we shall see that elements of $c_i(P), p_i(P), e(P)$ can be given by differential forms which is some polynomial in **F**, where **F** is a curvature 2-form of the bundle P on the manifold M.)

3 Connection and Curvature

The definition and notation in this part may be different from that in the next part since some notations used next is partly from physics. We will not fix this problem since some of them have been heavily used. However, this should be no ambiguity within each section.

We work with a principal $G-$bundle (E, π, M), where G is a Lie group, with Lie algebra $\mathfrak{g} = T_eG$, and Ad be the adjoint representation.

For a fixed point $x \in E$, the mapping $G \to E$ given by $g \mapsto xg, g \in G$, induces an injective map $v_x : \mathfrak{g} \to T_xE$ and the quotient space by the image of v_x is mapped to $T_{\pi(x)}M$ and is in fact isomorphic to it via the differential π_*. Namely, we have the following short exact sequence of vector spaces:

$$0 \to \mathfrak{g} \xrightarrow{v_x} T_xE \xrightarrow{\pi_*} T_{\pi(x)}M \to .$$

The vectors in the image of v_x are called *vertical tangent vectors*, and we want to choose a complement H_x of $v_x(\mathfrak{g})$, i.e., we want to find $H_x \subseteq T_xE$ such that $H_x \oplus v_x(\mathfrak{g}) = T_xE$. By the basic knowledge of short exact sequence, we know that this is equivalent to a splitting of the sequence, i.e., a map $\omega_x : T_xE \to \mathfrak{g}$ such that $\omega_x \circ v_x = \mathrm{id}_{\mathfrak{g}}$ and $H_x = \ker \omega_x$. This map ω_x should also vary differentially, i.e., $\omega \in \Omega^1(E, \mathfrak{g})$.

E.g. 3.1. We shall now give an important example, the Maurer-Cartan form. Consider a trivial bundle $E = M \times G, \pi : M \times G \to M$ and $\pi_2 : M \times G \to G$ be the natural projection, and $L_a : g \to G$ is the left translation by a. Then we define $\omega_{\mathrm{MC}} \in \Omega^1(E, \mathfrak{g})$ by

$$(\omega_{\mathrm{MC}})_{(p,g)} = (L_{g^{-1}} \circ \pi_2)_*, \qquad p \in M, g \in G.$$

Then it is not hard to show that $\omega_{\mathrm{MC}} = \pi_2^* \omega_0$, where $\omega_0 \in \Omega^1(G, \mathfrak{g})$ is defined by $(\omega_0)_g = (L_{g^{-1}})_*$. And it is also direct to see ω_0 is indeed differentiable, hence so is ω_{MC}. Besides, for $x = (p, g), v_x$ is the differential of the map $G \to M \times G$ given by $a \mapsto (p, ga)$, hence $(\omega_{\mathrm{MC}})_x \circ v_x = \mathrm{id}_* = \mathrm{id}$. So we see that this satisfies our original goal.

In fact, the form ω_0 on G is called the *Maurer-Cartan form.* For $G = \mathrm{GL}(n, \mathbb{R})$, it is just the form $\omega_0 = g^{-1} dg, g \in G$. It also satisfies a property that

$$R_g^* \omega_{\mathrm{MC}} = \mathrm{Ad}(g^{-1}) \circ \omega_{\mathrm{MC}},$$

where R_a is the right translation by a and $\mathrm{Ad}(g^{-1}) \circ : \Omega^1(E, \mathfrak{g}) \to \Omega^1(E, \mathfrak{g})$ is induced by the linear map $\mathrm{Ad}(g^{-1}) : \mathfrak{g} \to \mathfrak{g}$.

We would not provide the detail for the proof of the property above since it is not hard. Nevertheless, with this as motivation we could define general connections:

Def 3.1. A *connection* on a principal $G-$bundle (E, π, M) is a 1-form $\omega \in \Omega^1(E, \mathfrak{g})$ satisfying

(1) $\omega_x \circ v_x = \mathrm{id}$ where $v_x : \mathfrak{g} \to T_x E$ is the differential of the mapping $g \mapsto xg$;

(2) $R_g^* \omega = \mathrm{Ad}(g^{-1}) \circ \omega$ for all $g \in E$.

The second condition may seem abstract, however, using the complement space (which is called the *horizontal subspace*) given by ω satisfying the first condition as mentioned before, we may reformulate the second condition in an equivalent way:

Prop 3.1. For $\omega \in \Omega^1(E, \mathfrak{g})$ satisfying the first condition above, the second one is equivalent to

$$(R_g)_* H_x = H_{gx},$$

for all $x \in E, g \in G$. That is to say, the horizontal vector spaces are permuted by the right action of G on TE.

Hence by the discussion of the example above, ω_{MC} is a connection in $M \times G$ and is called the *Maurer-Cartan connection* or the *trivial connection*. Hence this gives the existence of a connection locally. Next we want to glue the local connection above to prove the general existence of connections on any principal G-bundle. This is given in the next proposition where we would also see the relation of connections between different bundles.

Prop 3.2. (1) Suppose $\omega_1, \cdots, \omega_k \in \Omega^1(E, \mathfrak{g})$ are connections on the bundle (E, π, M) and $\lambda_1, \cdots, \lambda_k \in C^\infty(M)$ satisfy $\sum_{i=1}^{k} \lambda_i = 1$, then $\omega = \sum_{i=1}^{k} \lambda_i \omega_i$ is also a connection on E.

(2) Let $(\tilde{f}, f) : (E', \pi', M') \to (E, \pi, M)$ be a bundle map of principal $G-$bundles and let $\omega \in \Omega^1(E, \mathfrak{g})$ be a connection in E, then $\tilde{f}^* \omega$ is a connection in E'.

(3) As a special case of (2), if $\varphi : F \to E$ is an isomorphism of principal $G-$bundles over M and $\omega \in \Omega^1(E, \mathfrak{g})$ is a conection in E, then $\omega^\varphi = \varphi^* \omega$ is a connection on F.

In our discussion of curvature forms, we would see they satisfy certain important properties whose definition we would single out here:

Def 3.2. (1) A k-form $\alpha \in \Omega^k(E, V)$ is called *horizontal* if $\alpha(X_1, \cdots, X_k) = 0$ for all $k-$tuples of tangent vectors $X_1, \cdots, X_k \in T_x E$ such that at least one of them is vertical.

(2) Let $\rho : G \to \mathrm{GL}(V)$ be a representation of G on V. Then $\alpha \in \Omega^k(E, V)$ is called $\rho-$*equivariant* if for all $g \in G$, we have $R_g^* \alpha = \rho(g^{-1})\alpha$.

Next we would define the curvature form associated to a connection. Since $\omega \in$

$\Omega^1(E,\mathfrak{g})$, we have $\omega\wedge\omega\in\Omega^2(E,\mathfrak{g}\otimes\mathfrak{g})$ and we may define $[\omega,\omega]\in\Omega^2(E,\mathfrak{g})$ to be the image of $\omega\wedge\omega$ by the linear mapping $[-,-]:\mathfrak{g}\otimes\mathfrak{g}\to\mathfrak{g}$ sending $X\otimes Y$ to $[X,Y]$.

Def 3.3. The *curvature form* $F_\omega\in\Omega^2(E,\mathfrak{g})$ for the connection ω is defined by the following structural equation

$$d\omega=-\frac{1}{2}[\omega,\omega]+F_\omega.$$

We summarize the most important properties of the curvature form in the following theorem:

Thm 3.1. (1) If $E=M\times G$ and $\omega=\omega_{\mathrm{MC}}$ is the Maurer-Cartan connection then $F_{\omega_{\mathrm{MC}}}=0$, that is $d\omega_{\mathrm{MC}}=-\frac{1}{2}[\omega_{\mathrm{MC}},\omega_{\mathrm{MC}}]$.

(2) In general F_ω is horizontal and Ad−equivariant, i.e., defines a 2-form $F_\omega\in\Omega^2(M,\mathfrak{g})$.

(3) (The Bianchi Identity) Further, $dF_\omega=[F_\omega,\omega]$. In particular, dF_ω vanishes on triples of horizontal vectors.

(4) Suppose $\{A_\alpha\}$ is a covering of M with trivializations $\{\varphi_\alpha\}_{\alpha\in\Sigma}$ of forms $\{A_\alpha\}_{\alpha\in\Sigma}$. Then the curvature form F_ω corresponds to the family $F_{A_\alpha}\in\Omega^2(U_\alpha,\mathfrak{g})$ given by $F_{A_\alpha}=dA_\alpha+\frac{1}{2}[A_\alpha,A_\alpha]$.

We would not prove (2), which is the most nontrivial one, whose proof would require deep consideration about one-parameter group associated to a vector field and hence too techinical to write out here. However, we would see how to use (2) to prove the rest of them and among them, (3) is the so-called Bianchi identity, whose equivalent form we would use in the following section.

Proof. We first notice that (1) follows from (2). In fact, by the discussion before, $\omega_{\mathrm{MC}}=\pi_2^*\omega_0$, where ω_0 can be viewed as the Maurer-Cartan connection on the bundle $G\to\mathrm{pt}$, that is, $F_{\omega_{\mathrm{MC}}}=\pi_2^*F_{\omega_0}$. But if by (2) we see F_{ω_0} is horizontal then it is 0, hence so is $F_{\omega_{\mathrm{MC}}}$.

Also (4) is directly from our definition. Hence what remains to be proved is the Bianchi identity.

We first differentiate the sturucal equation defining the curvature form:

$$\begin{aligned} 0 =& -\frac{1}{2}[d\omega, \omega] + \frac{1}{2}[\omega, d\omega] + dF_\omega = -[d\omega, \omega] + F_\omega \\ =& \frac{1}{2}[[\omega, \omega], \omega] - [F_\omega, \omega] + dF_\omega. \end{aligned}$$

Hence we only need to prove $[[\omega, \omega], \omega] = 0$, this comes from Jacobi identity, via the following computation:

$$\begin{aligned} &[[\omega, \omega], \omega](X, Y, Z) \\ &\quad =[[\omega, \omega](X, Y), \omega(Z)] - [[\omega, \omega](X, Z), \omega(Y)] + [[\omega, \omega](Y, Z), \omega(Z)] \\ &\quad =2([[\omega(X), \omega(Y)], \omega(Z)] - [[\omega(X), \omega(Z)], \omega(Y)] + [[\omega(Y), \omega(Z)], \omega(X)]) \\ &\quad =0 \end{aligned}$$

Hence the proof is finished. □

Def 3.4. A connection ω on a principal $G-$bundle is called *flat* if the curvature form vanishes everywhere, i.e., $F_\omega \equiv 0$.

In fact we have the following characterization of flat connection. Roughly speaking, locally they are just similar to the Maurer-Cartan connection introduced before and this explains why we would introduce them as an important example. However, we would not use this characterization later hence we would also omit its proof.

Thm 3.2. Let (E, π, M) be a principal $G-$bundle with connection ω. Then ω is flat if and only if around every point in M there is a neighborhood U with a trivialization $\varphi : \pi^{-1}(U) \to U \times G$ such that ω restricted to $E|_U$ is induce by Maurer-Cartan connection on $U \times G$, that is to say, $\omega|_{E|_U} = \varphi^* \omega_{\mathrm{MC}}$.

4 Chern-Weil Theory

As a continuation of the last section, we shall give some historical remarks now explaining the origin of the previous concepts. In particular this may explain why the notation in this chapter is somehow different.

Historically, people found in differential geometry the following curvature tensor $F^a_{\mu\nu}$:

$$F^a_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A^a_\mu + f_{abc} A^b_\mu A^c_\nu,$$

satisftying $[D_\mu, D_\nu] = -F^a_{\mu\nu} R_a$, where R_a is a structure constant.

Cartan introduce a map, which gives a curvature 2-form:

$$F^a_{\mu\nu} \mapsto \mathbf{F} = \frac{\lambda^a}{2i} \frac{F^a_{\mu\nu}}{2} dx^\mu \wedge dx^\nu.$$

And an insight is that $\mathbf{F} = d\mathbf{A} + \mathbf{A} \wedge \mathbf{A}$, where $\mathbf{A}$ is a connection on the bundle P which is a 1-form. And Cartan defined this to be $D\mathbf{A} := d\mathbf{A} + \mathbf{A} \wedge \mathbf{A}$, where D is a covariant exterior derivative. In fact, $D(D\mathbf{A}) = 0$, verifying the differential condition, which is a neat and somehow easy form of the Bianchi identity which also originates from differential geometry, with a messy appearance.

In fact, though different connections give different $\mathbf{F}$, we shall only consider those polynomials in $\mathbf{F}$ not depending on the connections $\mathbf{A}$ that determine the curvatures $\mathbf{F}$ (modulo an exact form). The independence comes from the fact that these polynomials are chosen to be the invariants of the Lie algebra $\mathfrak{g}$ corresponding to the Lie group G. The invariant of a Lie algebra is a polynomial in its generators $\{X_i\}$.

E.g. 4.1. For a compact Lie semisimple Lie algebra $\mathfrak{g}$, consider the Casimir element $X_1^2 + \cdots + X_n^2$, where $n = \dim G$ and the X_i are the canonical generators. They are in the center of the universal enveloping algebra of $\mathfrak{g}$ and are second order invariants in $\mathfrak{g}$.

From the Lie theory, we know that indeed all the invariants of a Lie algebra comes from the $m \times m-$minor of $\det(tI + a_i X_i)$, expanding it into a polynomial in t, namely, $\sum_{j=0}^{m} P_{m-j}(a_i)t^j$. It is easy to see that these polynomials $P_j(a_i)$ are invariant under the adjugate action of G. That is to see that $P(g^{-1}Xg) = P(X)$ and this is just direct to see from the invariance of determinant under similarity.

Consider $G = U(n)$. Then in fact the Chern class $c_j(P)$ is given by the following $\det(tI + \frac{i\mathbf{F}}{2\pi}) = \sum_j t^j P_{m-j}(\mathbf{F})$. We shall now prove the most basic thing: this can indeed represent a well-defined cohomology class in $H^{2j}(M, \mathbb{R})$, which is the work of Chern. This can be decomposed into two parts, the first is to prove that $P_j(\mathbf{F})$ is closed, which is somehow easier, while the second is to prove that $P_i(\mathbf{F})$ is independent from the choice of the connection $\mathbf{A}$, which is much harder and invovles heavier computation.

Let's first prove the first part. Viewing $P_j(\mathbf{F})$ as an invariant symmetric multilinear form in $j-$variables, namely, $P(x_1, \cdots, x_j)$, and $P(x, \cdots, x) = P_j(x)$. Then in fact, for any form $\mathbf{F}$, we have $dP_j(\mathbf{F}) = jP(D\mathbf{F}, \mathbf{F}, \cdots, \mathbf{F})$, where D is the covariant exterior derivative that $D\mathbf{A} = d\mathbf{A} + \mathbf{A} \wedge \mathbf{A} = \mathbf{F}$. We would not give the definition of the covariant exterior differential before, but here we need another form of Bianchi identity involving it later. Namely, $D(D\mathbf{A}) = 0$, and with this we see $D\mathbf{F} = 0$. Hence we are done. (In fact another thing that we assume to be true is the property of the covariant exterior derivative that we used at the first step in the computation. This would again be crucial in the next part.)

Then we are going to prove the second part. Then we need to prove that for two connections $\mathbf{A}_0, \mathbf{A}_1$ giving rise to curvatures $\mathbf{F}_0, \mathbf{F}_1$, then the difference $P_j(\mathbf{F}_0) - P_j(\mathbf{F}_1)$ is exact, that is, $= d\eta$ for a certain η. Consider the following family of connections $\mathbf{A}_t = \mathbf{A}_0 + t(\mathbf{A}_1 - \mathbf{A}_0)$ with curvature $\mathbf{F}_t$. Then note that $P_j(\mathbf{F}_1) - P_j(\mathbf{F}_2) = \int_0^1 \frac{d}{dt} P_j(\mathbf{F}_t)dt$. Then we only need to prove that $\frac{dP_j}{dt}(\mathbf{F}_t)$ is exact, since integral and d commutes, namely $\int_0^1 d\theta(t)dt = d(\int_0^1 \theta(t)dt)$. Note that by simple interval transformation from

$[t, 1]$ to $[0, 1]$ and change $\mathbf{A}_t$ by $\mathbf{A}_0$, we only need to prove (without loss of generality) that $\frac{dP_j}{dt}(\mathbf{F}_t)|_{t=0}$ is exact.

We proceed as in the first part: $\frac{dP_j}{dt}(\mathbf{F}_t) = jP(\frac{d\mathbf{F}_t}{dt}, \mathbf{F}_t, \cdots, \mathbf{F}_t)$. Denote $\mathbf{A}_1 - \mathbf{A}_0$ by $\mathbf{a}$, then

$$\begin{aligned}\mathbf{F}_t =& d\mathbf{A}_t + \mathbf{A}_t \wedge \mathbf{A}_t = d(\mathbf{A} + t\mathbf{a}) + (\mathbf{A} + t\mathbf{a}) \wedge (\mathbf{A} + t\mathbf{a}) \\ =& \mathbf{F} + t(d\mathbf{a} + \mathbf{A} \wedge \mathbf{a} + \mathbf{a} \wedge \mathbf{A}) + t^2 \mathbf{a} \wedge \mathbf{a}.\end{aligned}$$

Hence $\frac{d}{dt}(\mathbf{F}_t)|_{t=0} = d\mathbf{a} + \mathbf{A} \wedge \mathbf{a} + \mathbf{a} \wedge \mathbf{A}$. In fact this is just $D\mathbf{a}$. (Since for a matrix valued $p-$form ω under the action of G, we have the formula $D\omega = d\omega + \mathbf{A} \wedge \omega + (-1)^{p+1}\omega \wedge \mathbf{A}$. Here $\mathbf{a}$ is understood as a matrix valued 1-form.) Then finally, $\frac{dP_j}{dt}(\mathbf{F}_t)|_{t=0} = jP(D\mathbf{a}, \mathbf{F}, \cdots, \mathbf{F}) = d(jP(\mathbf{a}, \mathbf{F}, \cdots, \mathbf{F}))$ by the very same property of the covariant exterior derivative that we've used and we are done.

Next we turn to consider the Понтрягин class, using the same manner. The group $G = O(n)$. Now we shall prove calculate $\det(tI - \frac{\mathbf{F}}{2\pi})$. First a useful obeservation is that

Prop 4.1. If $\mathbf{F}$ is $\mathfrak{o}(n)-$valued, then $P_j(\mathbf{F}) = 0$ for all odd numbers j.

Proof. Since $\mathbf{F} \in \mathfrak{o}(n)$, namely $\mathbf{F}^t = -\mathbf{F}$, then $\det(tI - \frac{\mathbf{F}}{2\pi}) = \det((tI - \frac{F}{2\pi})^t) = \det(tI + \frac{\mathbf{F}}{2\pi})$. Then we see directly by expanding this formula and comparing the coefficients. □

As a corollary, Понтрягин class are always $4i-$forms, since $\mathbf{F}$ is a $2-$form, and hence we denote $P_j(P) \in H^{4j}(M, \mathbb{R})$.

(Similarly, for $k = 2m$, $G = SO(k)$, an elemnt in $H^k(M; \mathbb{R}) = f^* H^k(\widetilde{\mathrm{Gr}}(n, k, \mathbb{R}); \mathbb{R})$ is called Euler class, denoted as $e(P)$. For $k = 2m$, $e(P)$ is a $2m-$form, hence $e(P) \wedge e(P)$ is a $4m-$form. Hence if we take any element in Понтрягин class that is a $4m-$form $p_m(P), m = \frac{k}{2}$, then the Euler class is defined to be the $k-$form satisfying $e(P) \wedge e(P) = p_{\frac{k}{2}}(P)$.)

Finally, we discuss Euler class. The group $G = SO(n)$ and the associated Lie algebra is still $\mathfrak{o}$. Recall the definition of a Euler class: the class $e(P)$ satisfying that $e(P) \wedge e(P) = p_m(P) = P_{2m}(\mathbf{F})$, but such $e(P)$ is not uniquely determined. For example, the sign can be both positive or negative. To make it well-defined, we shall use the Pfaffian. Recall the definition of Pfaffian. In fact, the determinant of a $2m \times 2m$ skew-symmetric matrix is a square of a certain polynomial (or its negative). The one with positive leading coefficient is called the Pfaffian of this matrix.

Then to avoid different choices for the sign, we just define $e(P) = \frac{1}{(2\pi)^m}\mathrm{Pf}(\mathbf{F})$. Since the Pfaffian is $SO(k)-$invariant (but not $O(k)-$invariant), we see directly that $e(P)$ is $SO(k)-$invariant. And in particular, when $k = 2m$, this is an exceptional characteristic class obtained from the restriction from $O(k)$ to $SO(k)$.

Before we start next to do some concrete calculations, we need some preparation. Note that we've only defined characteristic classes for principal bundle and now we shall generalize to any vector bundle. We shall define a universal bundle, but with vector space as fiber, not groups. In other words, for a vector bundle ξ, we want to define the universal bundle ξ_E that classifies real vector bundles of rank E, which has the same base space with ξ but different fibers.

Def 4.1. Let E be a real vector bundle of rank k over manifold M whose dimension is d, with structure group $\mathrm{GL}(k, \mathbb{R})$. Without loss of generality, we can reduce it to $O(k)$. Then we define the *universal bundle* $\xi_E(n-k-1, \mathbb{R}^k)$ to be a real vector bundle of rank k over the base space $\mathrm{Gr}(n, k, \mathbb{R})$. Its fiber over a point $x \in \mathrm{Gr}(n, k, \mathbb{R})$ is just the subspace corresponding to x (just tautologically).

When E is a complex vector bundle of rank k with structure group $\mathrm{GL}(k, \mathbb{C})$. Without loss of generality, we can reduce it to $U(k)$. Then we define the *universal bundle* $\xi_E(2n-2k, \mathbb{C}^k)$ to be a real vector bundle of rank k over the base space $\mathrm{Gr}(n, k, \mathbb{C})$. Again, its fiber over a point $x \in \mathrm{Gr}(n, k, \mathbb{C})$ is the subspace corresponding to x.

As in the case of principal bundle, here $\xi_E(n-k-1, \mathbb{R}^k)$ classifies all the real vector bundles of rank E. Namely, if $d \leq n-k-1$ and $f : M \to \mathrm{Gr}(n,k,\mathbb{R})$, then every such bundle E on M comes from $f^*\xi_E(n-k-1, \mathbb{R}^k)$. The story in complex case is analogous.

Now let's see some computation examples. First we calculate some Chern classes.

Let P be a principal bundle over manifold M with structure group $SU(2)$ whose Lie algebra is $\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) | \mathrm{tr}\,(X) = 0, X^* = -X\}$. The its standard basis of Pauli matrices is

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

From an example from physics, we shall consider a curvature $\mathbf{F} = \sum_a \mathbf{F}_a \frac{\sigma_a}{2i}$, where $a = 1, 2, 3$. Then we calculate

$$\begin{aligned} \det(tI + \frac{i}{2\pi}\mathbf{F}) =& \det(tI + \frac{1}{4\pi}\sum_a \mathbf{F}_a\sigma_a) \\ =& t^2 + \frac{t}{2\pi}\mathrm{tr}\,(\sum_a \mathbf{F}_a\sigma_a) + \frac{1}{32\pi^2}\sum_{a,b}(\mathrm{tr}(\mathbf{F}_a\sigma_a)\mathrm{tr}\,(\mathbf{F}_b\sigma_b) - \mathrm{tr}(\mathbf{F}_a\mathbf{F}_b\sigma_a\sigma_b)). \end{aligned}$$

(I do not use the notation inherited from physics here since the summation convention is horrible for me.) The constant term here comes from nothing but the identity that $\det(A) = \frac{1}{2}(\mathrm{tr}\,(A)^2 - \mathrm{tr}(A^2))$, which can easily seen by characteristic values. Also, since Pauli matrices $\sigma_a \in \mathfrak{su}(2), \mathrm{tr}\,\sigma_a = 0$ we see the coefficient of t is 0. Also, we can cehck that $\mathrm{tr}(\sigma_a\sigma_b) = 2\delta_{ab}$, hence $\det(tI + \frac{i\mathbf{F}}{2\pi}) = t^2 - \frac{\sum_a \mathbf{F}_a^2}{16\pi^2}$. Denote by Chern classes and we see that the result just means that $c_1(P) = 0, c_2(P) = -\frac{1}{16\pi^2}\sum_a \mathbf{F}_a^2$.

Rmk 4.1. For historical remark, $SU(2)$ is the somehow easiest nonabelian Lie group, and several physicians tried to generalize the usual Maxwell equations (for the $U(1)-$bundles)

to $SU(2)-$bundles. In fact, if we write

$$\begin{aligned} E =& E_x dt \wedge dx + E_y dt \wedge dy + E_z dt \wedge dz \\ B =& B_x dz \wedge dy + B_y dx \wedge dz + B_z dy \wedge dx \\ F =& E(\vec{E}) + B(\vec{B}). \end{aligned}$$

Then the (simplified version of) Maxwell equation just reads $dF = d * F = 0$. When considering such things for $SU(2)-$bundles, Pauli did similar calculation and obtained $-\frac{1}{16\pi^2}\sum_a \mathbf{F}_a^2$, whose physical meaning took a long time to be understood.

Next we turn to an example of computation of Понтрягин classes. Here M is a Riemann manifold with dimension 2 and we consider its tangent bundle. We have seen before the only nonvanishing Понтрягин classes have order divisible by 4, hence here we only need to calculate $p_0(TM) = 1$ and the higher classes vanish.

But as a by-product, we can also calculate the Euler classes $e(TM)$, which satisfies $e_1(TM) \wedge e_1(TM) = p_1(M) = 0$. But as we shall next see, it is not 0 itself. In fact, y the Pfaffian definition, $e_1(TM) = \frac{1}{2\pi}\mathrm{Pf}(\mathbf{F}) = -\frac{1}{4\pi^2}\epsilon_{\alpha\beta}\mathbf{F}_\beta^\alpha \neq 0, \alpha, \beta \in 1, 2$.

What if M is of dimension 4? We calculate the Понтрягин classes and Euler classes of its tangent bundle. As before, p_2 and higher classes is just 0 by dimension argument. And we can calculate p_1 by the same method. Namely, we take $\mathbf{F}$ to be a $\mathfrak{o}(4)-$valued 2-form, then $\det(tI - \frac{\mathbf{F}}{2\pi}) = t^4 + p_1(TM)t^2 + p_2(TM)$ and analogously we get $p_1(TM) = -\frac{1}{8\pi^2}\mathrm{tr}(\mathbf{R} \wedge \mathbf{R})$ (where we change $\mathbf{F}$ by $\mathbf{R}$ to view it as a Riemannian curvature tensor). Also, using Pfaffian, we have $e(TM) = \epsilon_{\alpha\beta\mu\nu}\mathbf{R}_\beta^\alpha \wedge \mathbf{R}_\nu^\mu$. In particular, $e, p_1 \neq 0$ but $p_2 = 0$.

5 Global Invariant

Appropriate characteristic classes will give us some global invariants of a manifold, manifesting its structure. We shall see some examples below.

First, we will see some invariants about the differentiable structure of M. In particular, we shall prove that there exist different differentiable structures on $\mathbb{S}^7$, that is to say, it is not in general true for a manifold homeomorphic to $\mathbb{S}^7$ to be diffeomorphic to $\mathbb{S}^7$. In fact, there are 28 diffeomorphic equivalent classes.

The basic idea is that if M, M' are diffeomorphic, then $p_i(M) = p_i(M')$. But for those homeomorphic but not diffeomorphic M, M', we have $p_i(M) \neq p_i(M')$. We need the following theorem to obtain two homeomorphic 7-dimensional manifolds. In particular it shows that if M is the boundary of an 8-dimensional manifold E, then M is homeomorphic to $\mathbb{S}^7$.

Thm 5.1 (Reeb)**.** If a compact differentiable manifold X of dimension n admits a differentiable function $X \to \mathbb{R}$ with exactly two critical points which are non-degenerate, then X is homeomorphic to an n-sphere (with its Euclidean metric topology).

Also, we would need the sign formula by Hirzebruch-Thom:

$$\text{sign}(M) = \frac{1}{45}(7p_2 - p_1^2)[M],$$

where M is a manifold of dimension 8 and $\text{sign}(M)$ is 1 if and only if M is orientable and -1 otherwise. Then if we take an orientable M, in particular we see that $p_2[M] \neq 0$ but $p_2(\mathbb{S}^7)$ is indeed 0, hence we've found an example.

Next, we would consider some invariants about the complex structure of M. Consider an Euler form $e(\mathbf{F})$, then we have the following result:

$$\int_M e(\mathbf{F}) = \chi(M),$$

where $\chi(M)$ is the Euler characteristic number.

This formula dates back to the famous Gauss-Bonnet theorem:

Thm 5.2. For a surface M, if ∂M is piecewise smooth, then

$$\sum_{i=1}^{n}(\pi - \alpha_i) + \int_{\partial M} k_g ds + \int_M k dA = 2\pi\chi(M).$$

We shall omit the description of the terms in this theorem, since they are just different kinds of curvatures so the theorem is connecting the curvature which is defined locally to a global invariant. And in particular, for a 2 dimensional manifold without boundary, the formula reduces to

$$\int_M k dA = 2\pi\chi(M).$$

Now we turn to the more general formula above. In fact, there is a result (which we again use without proof) that for a complex $n-$dimensional manifold, if we view the tangent bundle TM as a real bundle of rank $2n$, then $p_n(TM_{\mathbb{R}}) = c_n^2(TM) = (c_n(TM) \wedge c_n(TM))$. Then recall that by definition $p_n = e \wedge e$, the integrated term in the formula can be altered to be c_n. Namely, we have the following:

$$\int_M c_n(\mathbf{F}) = \chi(M).$$